

Run 06 | NME

India

IV only

Target Necessity-motivated entrepreneurship; binary 1 vs. 0
Sample 27,555 respondents **Coverage** 1 country; 2015–2022
Positive cases 1,849 (6.71%) **Case time** 1m 02s
Train / val. / test 18,740 / 3,305 / 5,510 (68% / 12% / 20%, stratified)
Predictors (13): Self-efficacy, opportunity perception, fear of failure, knowing entrepreneurs, intention, media coverage, business exit, business angel, established business owner, good career choice, ease of starting a business, high status, social enterprise.
Fraser EFW predictors (7): Overall economic freedom, Economic freedom quartile, Size of government, Legal system and property rights, Sound money, Freedom to trade internationally, Regulation. Total independent variables: 20.
Controls (0): None. Country and respondent ID are excluded as predictors.
Protocol: Ten CPU models; model seed 20260925; fixed splits by sample/outcome; training-only preprocessing; F1 threshold selected on validation data. Test scores below are unweighted.

Result in brief

Highest observed ROC-AUC: H2O GBM (0.9886). Highest F1: CatBoost (0.7878). The all-negative accuracy baseline is 93.29% (full sample).

Model performance on held-out test records

Model	ROC-AUC	PR-AUC	Acc.%	Prec.%	Recall%	F1	Time
H2O GBM	0.9886	0.8329	96.61	70.68	84.28	0.7689	3s
XGBoost	0.9883	0.8262	96.97	76.03	79.95	0.7794	<1s
CatBoost	0.9880	0.8186	97.10	77.14	80.49	0.7878	13s
Random Forest	0.9872	0.8122	96.90	75.00	80.49	0.7765	1s
Elastic Net	0.9855	0.7606	96.24	67.92	83.20	0.7479	16s
Lasso	0.9855	0.7606	96.24	67.92	83.20	0.7479	12s
Decision Tree	0.9853	0.7494	96.88	74.69	80.76	0.7760	<1s
Logistic Regression	0.9853	0.7578	96.08	65.39	88.08	0.7506	<1s
Ridge	0.9845	0.7642	95.86	64.02	87.26	0.7385	9s
Naive Bayes	0.8978	0.4508	93.09	48.77	64.23	0.5544	5s

Models ranked by ROC-AUC; small differences are descriptive, not significance tests. Time includes fit, prediction and output processing. Logistic regression uses a small stabilizing penalty; Lasso/Ridge/Elastic Net use 3-fold training CV.

Random-forest feature importance: top 8

Rank	Predictor	Raw importance	Share%
1	Entrepreneurial intention	711.54	48.44
2	Established business owner	277.53	18.90
3	Knowing entrepreneurs	36.08	2.46
4	Sound money	34.26	2.33
5	Freedom to trade internationally	28.84	1.96
6	Size of government	27.89	1.90
7	Legal system and property rights	27.06	1.84
8	Self-efficacy	24.33	1.66

Share is the percentage of total forest impurity importance across all 38 encoded predictors, not explained variance. Raw values are not comparable across samples. Missing indicators are derived preprocessing features.

Leading Fraser EFW predictors within the same forest

Rank	Fraser EFW predictor	Raw importance	Share%
1	Sound money	34.26	2.33
2	Freedom to trade internationally	28.84	1.96
3	Size of government	27.89	1.90

Fraser EFW subset ranked by raw importance; shares use total forest importance, not the Fraser EFW subtotal.

Source: Completed Objective 3 batch, 29 September 2026; 06_obj3_india_iv_NME. Scores and importance are taken directly from saved CSV outputs.